**Jacobian of Alpoge's candidate counterexample map** $F : \mathbb{C}^3 \to \mathbb{C}^3$

$$F(x,y,z) \;=\; \Big( (1+xy)^3 z + y^2(1+xy)(4+3xy), \quad y + 3x(1+xy)^2 z + 3xy^2(4+3xy), \quad 2x - 3x^2 y - x^3 z \Big)$$

$$\frac{\partial F}{\partial(x,y,z)} \;=\; \left(\begin{matrix} 3y^3\left(xy+1\right) + y^3\left(3xy+4\right) + 3yz\left(xy+1\right)^2 & 3xy^2\left(xy+1\right) + xy^2\left(3xy+4\right) + 3xz\left(xy+1\right)^2 + 2y\left(xy+1\right)\left(3xy+4\right) & \left(xy+1\right)^3 \\ 9xy^3 + 6xyz\left(xy+1\right) + 3y^2\left(3xy+4\right) + 3z\left(xy+1\right)^2 & 9x^2y^2 + 6x^2z\left(xy+1\right) + 6xy\left(3xy+4\right) + 1 & 3x\left(xy+1\right)^2 \\ -3x^2z - 6xy + 2 & -3x^2 & -x^3 \end{matrix}\right)$$

$$\det\!\left(\frac{\partial F}{\partial(x,y,z)}\right) \;=\; -2 \qquad \text{(SymPy-verified identity in } \mathbb{Q}[x,y,z]\text{)}$$

Three distinct points collapse to the same image:

$$F\big(0,\,0,\,-\tfrac{1}{4}\big) \;=\; F\big(1,\,-\tfrac{3}{2},\,\tfrac{13}{2}\big) \;=\; F\big(-1,\,\tfrac{3}{2},\,\tfrac{13}{2}\big) \;=\; \big(-\tfrac{1}{4},\,0,\,0\big)$$

Map and points due to Levent Alpoge (`@__alpoge__`), announcement 2026-07-20.
This document is a symbolic re-derivation only.